

TECHNICAL DOCUMENTATION: OMNICOZY MOBILE SHADER

Version: 2.0 (2026 Edition) **Author:** PixeliusVita **Pipeline:** Universal Render Pipeline (URP)

1. PRODUCT OVERVIEW

The **OmniCozy Mobile Shader** is a high-performance "Uber Shader" specifically engineered for indie developers creating stylized, cozy, and casual games. It provides a tactile, "marshmallow-soft" aesthetic by simulating complex lighting effects—such as Global Illumination and Subsurface Scattering—within a single, mobile-optimized pass.

2. CORE FEATURES & MODULES

A. Tonal Harmony & Color Grading

- **Scene Unification:** Allows for a global color cast to be applied to all materials to ensure scene-wide color harmony.
- **Built-in Post-Processing:** Features integrated Saturation, Contrast, and Desaturation controls to reduce the need for expensive screen-space post-processing.

B. Stylized "Clay" Lighting

- **Light Wrap:** Simulates light bleeding into shadow areas, giving models a soft, "clay-like" feel.
- **Highlight Roll-off:** Prevents HDR light "burning" on stylized surfaces, maintaining pastel color integrity.
- **Shadow Lift:** Prevents dark crevices from crushing to pure black, keeping the look soft and inviting.

C. Volume & Ambient Simulation (GI)

- **Fake Global Illumination:** Uses a hemisphere lighting model (Sky/Ground colors) to give objects volume even in unlit areas.
- **Tinted SSAO:** Tints ambient occlusion crevices with color instead of black for a cleaner, artistic look.
- **Procedural Bottom AO:** Automatically generates grounding shadows based on local position to anchor objects to the floor.

D. Material Effects & Outlines

- **Elemental Toggle:** Supports stylized metallic and smoothness effects for reflective surfaces.
- **Glass Transparency:** Optimized alpha-blending for translucent props.
- **Inverted Hull Outlines:** Clean, consistent silhouettes with customizable tinting and width.

3. TECHNICAL SPECIFICATIONS

- **Target Platforms:** Mobile (iOS/Android), PC, Console.
 - **Workflow:** Optimized for **Texture Palette** workflows to achieve ultra-low Draw Calls.
 - **Shader Keywords:** Uses `shader_feature_local` to ensure that only active features are compiled, stripping away unused code to maximize FPS.
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4. HOW TO USE

1. **Assign Shader:** Apply `PVFX/PixeliusVita_OmniCozyMobileShader` to your material.
2. **Toggle Features:** Enable only the modules required (e.g., Enable Outline, Enable Metal).
3. **Unified Look:** Use the **Harmony Intensity** and **Tonal Tint** to match the material to your scene's global lighting mood.